



COURSE DESCRIPTION FORM

INSTITUTION FAST University of Computer and Emerging Sciences

PROGRAM (S) TO BE BS(DS)

EVALUATED

A. Course Description

(Fill out the following table for each course in your computer science curriculum. A filled out form should not be more than 2-3 pages.)

Course Code	DS2001
Course Title	Introduction to Data Science
Credit Hours	3
Prerequisites by Course(s) and Topics	-
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	Assignments: 10 Quizzes: 10 Project: 10 Mid Term I: 12.5 Mid Term II: 12.5 Final: 45
Course Coordinator	Dr. Muhammad Nouman Durrani
URL (if any)	
Current Catalog Description	Data Science is a dynamic and fast-growing field at the interface of Statistics and Computer Science. It is an interdisciplinary field about processes and systems to extract knowledge or insights from data in various forms. This course will introduce students to this rapidly growing field and equip them with some of its basic principles and tools including data collection and integration, data cleaning, data analysis using machine learning, visualization and effective communication. The main focus of these topics will be on understanding and integration of concepts and their application to solving problems.
Textbook (or Laboratory Manual for Laboratory Courses)	Lecture Notes Davy Cielen, Arno D. B. Meysman, and Mohamed Ali, Introducing Data Science, Big data, machine learning, and more, using Python tools, May 2016
Reference Material	Journals: Machine Learning, Pattern Recognition

Program Learning Outcomes	Conferences: ICPR, ICDM, ICML, KDD www.datacamp.com (Students are given free access on many tutorials)			
	PLO1	Computing Knowledge	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems.	✓
	PLO2	Problem Analysis	Identify, formulate, research literature, and analyse complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.	✓
	PLO3	Design/Develop Solutions	Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	✓
	PLO4	Investigation & Experimentation	Conduct investigation of complex computing problems using research based knowledge and research based methods	✓
	PLO5	Modern Tool Usage	Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modelling for complex computing problems.	✓
	PLO6	Society Responsibility	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to context of complex computing problems..	✓
	PLO7	Environment and Sustainability	Understand and evaluate sustainability and impact of professional computing work in the solution of complex computing problems	✓
	PLO8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of computing practice.	✓
	PLO9	Individual and Team Work	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.	✓
	PLO10	Communication	Communicate effectively on complex computing activities with the computing community and with society at large.	✓
	PLO11	Project Management and Finance	Demonstrate knowledge and understanding of management principles and economic decision making and apply these to one's own work as a member or a team.	✓
	PLO12	Life Learning Long	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes.	✓

Course Goals	Course Learning Objectives: CLO 1: Students should be able to articulate a comprehensive understanding of Data Science, encompassing its fundamental concepts, methodologies, and practical applications. CLO 2: Students should be able to design and implement data pre-processing pipelines to clean, transform, and prepare raw data for analysis, ensuring data quality and integrity. CLO 3: Students should be able to apply programming languages and open-source tools to manipulate data, develop predictive models, and visualize results effectively. CLO 4: Students should be able to evaluate and select appropriate algorithms and tools for specific analytical tasks, considering factors such as scalability, accuracy, and interpretability of results. CLO 5: Students should be able to work as a team while integrating important components in data science <i>Student Outcomes Addressed by the Course (From ABET)</i> (a) An ability to apply knowledge of computing and mathematics appropriate to the program's student outcomes and to the discipline (b) An ability to analyze a problem, and identify and define the computing requirements appropriate to its solution (f) An ability to communicate effectively with a range of audiences (j) An ability to apply mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems in a way that demonstrates comprehension of the tradeoffs involved in design choices [CS] (k) An ability to apply design and development principles in the construction of software systems of varying complexity [CS]																								
Relation between CLOs and PLOs	<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 15%;">CLO No.</th><th style="width: 20%;">Domain</th><th style="width: 20%;">Taxonomy level</th><th style="width: 15%;">PLO</th></tr> </thead> <tbody> <tr> <td>1</td><td>Cognitive</td><td>3</td><td>1</td></tr> <tr> <td>2</td><td>Cognitive</td><td>4</td><td>2</td></tr> <tr> <td>3</td><td>Cognitive</td><td>5</td><td>2</td></tr> <tr> <td>4</td><td>Cognitive</td><td>4</td><td>5</td></tr> <tr> <td>5</td><td>Cognitive</td><td>5</td><td>9</td></tr> </tbody> </table>	CLO No.	Domain	Taxonomy level	PLO	1	Cognitive	3	1	2	Cognitive	4	2	3	Cognitive	5	2	4	Cognitive	4	5	5	Cognitive	5	9
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4	Cognitive	4	5																						
5	Cognitive	5	9																						

Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)	1. Topics to be covered:			
	List of Topics	Week	Contact Hours	CLO
	Basics of Data Science, Big data, Types of Data,) Data Science, Motivating Examples, Data Science process	1	2	1, 4
	Data Science Lifecycle /,Data Types and sources (Kaggle, UCI, APIs)/Data acquisition methods (APIs, scraping basics)/Detecting missing values and outliers/Remove duplicates	2	1	4
	Data Preparation: Data Cleaning EDA: Compute Simple Statistics, Simple Visualization, Exploratory Data Analysis with Python/Summary statistics, shape, skew, kurtosis/Data Visualization with Python/istograms, boxplots, scatter plots/Correlation matrix and heatmap/Create visualizations using Matplotlib and Seaborn/Visual storytelling	3	2	1, 2, 4
	Intro to ML/Train/test split, cross-validation/Basic ML pipeline/ Model Evaluation Metrics /Accuracy, Precision, Recall, F1 Score/ CrossValidation/Confusion Matrix/ROC/AUC Curves/Bias-variance tradeoff, Intro to Supervised Learning start	4	1	4
	Supervised Learning(Regression)/ Linear Regression/ Cost function and Gradient Descent/Underfitting vs Overfitting/Cross validation	5	3 3	3, 4 3, 4
Mid 1 exam (6th week)				

	Supervised Learning(Classification) / Logistic Regression/ k-Nearest Neighbors (k-NN)/Decision Trees and Random Forests	7	6	3, 4
	Unsupervised Learning – Clustering/ K-Means Clustering(Elbow method, centroid initialization)/Hierarchical Clustering	8	9	3, 4
	Mid II exam (12th week)			
	Feature Extraction & Dimensionality Reduction /Singular Value Decomposition (SVD)/Linear Discriminant Analysis (LDA) – contrast with PCA	9	3	2, 3, 4
	Associative Learning /Market Basket Analysis/Apriori algorithm: support, confidence, lift/PCY algorithm & multi-hashing for large datasets	10	1	4
	Recommendation Systems, Ensembles & Streaming Data, Recommendation Systems/User-based & Item-based Collaborative Filtering/Cosine/Pearson similarity/ Matrix factorization (brief overview) Ensemble Learning/Bagging vs Boosting/Random Forest, AdaBoost, Gradient Boosting	11,12	3	2, 3, 4
	Data Science Ethics & Applied Data Science Capstone/ Bias in ML models/ Data privacy, surveillance, algorithmic fairness/ /Peer feedback and summary discussion	13	3	3, 4
	Advance Applications in Data Science/ Google PageRank & Intro to Graph Analytics / PageRank intuition & formula/ Graph representation & traversal/ Applications in networks, social media/ Hadoop MapReduce: Intro and programming flow/	14,15	3	2, 3, 4
	Project Presentations	16	3	5
	Total	16	48	



Instructor Name : Mahrukh Khan

Date: 20-08-2025



Laboratory Projects/Experiments Done in the Course	Yes. Project and Regular Lab Classes once a week			
Programming Assignments Done in the Course	Yes. All topics were practically covered through programming assignments.			
Class Time Spent on (in credit hours)	Theory	Problem Analysis	Solution Design	Social and Ethical Issues
	40 %	25 %	25%	10%
Oral and Written Communications	Every student is required to submit at least __1__ written reports of typically __8__ pages and to make __1__ oral presentations of typically __10__ minute's duration. Include only material that is graded for grammar, spelling, style, and so forth, as well as for technical content, completeness, and accuracy.			

Instructor Name Sania Urooj

Instructor Signature _____

Date 17/01/2025